

Table 1
Test substrate parameters.

Device no.	$d_{p,nom}$ (μm)	T_{anneal} ($^{\circ}\text{C}$)	t_{anneal} (min)	$R_{a,low}$ (μm)	$R_{a,high}$ (μm)	$R_{q,low}$ (μm)	$R_{q,high}$ (μm)
1	–	–	–	0.263	0.341	0.319	0.423
2	15	725	5	0.622	0.726	0.789	0.930
3	30	740	15	1.396	1.649	1.755	2.09
4	100	725	5	4.25	5.17	5.22	6.66
5	250	725	5	7.51	13.04	9.13	16.46
6	100	740	15	4.40	5.48	5.41	7.16
7	100	750	45	3.73	5.04	4.58	6.38

successively onto the ITO surface to provide for isopotential electrical contacts.

2.3. Patterning

The parts were then subjected to a 2-step conventional microlithography process in order to pattern ITO heater strips and gold contacts. The photolithography mask patterns are shown in Fig. 2. Before each of the two steps, the test pieces were cleaned in toluene, acetone, and isopropanol for 5 min each, then prebaked for 10 min in a 90 $^{\circ}\text{C}$ oven to drive off residual liquid molecules. AZ-9260 (AZ Electronic Materials USA Corp.) positive photoresist (PR) was applied to the surface by spinning at 4000 rpm for 40 s, with a 2-s ramp up in order to prevent the relatively massive test piece from slipping off the spinner vacuum chuck. The resulting PR thickness for the spin recipe was approximately 7 μm [27], enough to coat even the largest asperities (as verified by visual inspection under a microscope). The PR-coated parts were softbaked in a 110 $^{\circ}\text{C}$ oven for 4 min, then inserted into the mask aligner (Karl Suss model MJB-3). After aligning the mask to the part, the coated surface was subjected to UV light from the aligner’s internal mercury arc lamp at an intensity of 1200 mJ/cm^2 for 36 s. Following exposure, the test pieces were developed in a 3:1 solution of water and AZ 400 K for 2 min, and then hardbaked in a 90 $^{\circ}\text{C}$ oven for 10 min.

In the first patterning step, the entire heater and marker array was patterned by successively etching the gold, titanium, and ITO layers, removing all areas not appearing black in Fig. 2a. In the second patterning step, the photoresist was patterned according to the black areas of Fig. 2b. Only gold and titanium were removed in this step, leaving the ITO pattern unmodified.

The 200-nm gold layer was removed in approximately 2 min by immersion in a gold-specific etchant containing potassium iodate (Transene Gold Etch TFA). The 10-nm titanium adhesion layer was then etched using “base piranha” solution consisting of three parts $\text{NH}_4(\text{OH})$ (29% by mass in water) to 1 part H_2O_2 (30% by mass in water), requiring about 30 s for complete removal. The ITO layer was etched in concentrated HCl (37% by mass in water), a process that required soaking between 5 and 15 min depending upon the ITO coating batch. Undercutting of up to 10 μm was observed, the HCl etch being anisotropic and local etch rates being highly dependent upon the material composition and/or surface geometry. After etching, the photoresist etch stop was removed by soaking in toluene, followed by acetone and then isopropanol.

Completed parts (see Fig. 3) contained four test strips of different widths flanked by gold electrodes with good adhesion and electrical contact properties, with minimum separations of 1 mm between electrical contacts for different strips. The test strips chosen for this study measured 25 mm long $\times$ 400 μm wide, a width which was found experimentally to provide an optimum combination of heat transfer data and bubble visualization. Each completed part also contained a 25 mm long $\times$ 16 mm wide test area so that sessile drop contact angle measurements could be performed on a large ITO-coated area.